Executive Summary

Factory Reallocation & Shipping Optimization Recommendation System

Prepared as an analytical prototype and decision-support proposal · October 2026

Purpose

A decision-support prototype to compare factory assignments and shipping scenarios, with a roadmap for predictive lead-time modeling and constrained optimization.

Current baseline

10,194 order lines; 8,549 orders; 15 products; $141,783.63 sales; $93,442.80 gross profit; 65.91% weighted gross margin. Chocolate contributes 92.88% of sales.

Prototype capabilities

- Compare five provided factory coordinates for selected product and state proxy.
- Compare configured current factory against alternatives.
- Rank alternatives using a speed-versus-proxy-cost slider.
- Show risks, missing mapping, and data-readiness gaps.

Safeguards

- Median order-to-ship gap is 1,274.0 days and needs validation.
- Actual factory assignment history, freight costs, capacity, and destination coordinates are absent.
- Displayed shipping time/cost are illustrative proxies, not trained predictions or actual savings.
- Human review is required before any operational decision.

Recommended next steps

- Validate date fields and target definition.
- Collect actual factory, delivery, cost, capacity, and compatibility data.
- Train and evaluate regression candidates with MAE, RMSE, and R^2 after data validation.
- Back-test and pilot with human approval.

Decision requested

Approve a data-validation and pilot-design phase, not immediate factory reassignment. Do not claim operational savings until actual shipment and cost data validate the recommendations.